SANDRA SAWDIATU INUSAH

Geospatial Analysis • Geospatial AI • Land Use and Human–Environment Interaction
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Research Profile

I am a graduate geospatial researcher with over four years of experience using remote sensing, geospatial technology, machine learning, and environmental data to study land-use and land-cover change, vegetation dynamics, and human–environment interactions. I have experience integrating multi-source Earth observation datasets, including Landsat, UAV imagery, GEDI LiDAR, and field measurements, to investigate environmental changes, and I am skilled in Python, R, Google Earth Engine, spatial modeling, machine learning, and satellite time-series analysis. Current MSc research focuses on mapping land-use transitions and vegetation structural changes across Ghana's forest–savanna transition zone using remote sensing and machine-learning approaches.

Research interests: geospatial artificial intelligence; earth observation foundation models; satellite time-series analysis; UAV–satellite data integration; land-system science; scalable geospatial methods for environmental monitoring.

Education

MSc, Geography 2025 – 2027 (expected)
University of British Columbia *Vancouver, Canada*

- Thesis: *Spatiotemporal Analysis of Vegetation Dynamics in Ghana's Forest–Savanna Transition Zone Using Multi-Source Remote Sensing and Machine Learning*
- Supervisor: Dr. Naomi Schwartz • Co-supervisor: Dr. Nicholas Coops

BSc, Natural Resources Management (First Class Honours) 2018 – 2022
Kwame Nkrumah University of Science and Technology (KNUST) *Kumasi, Ghana*

- Specialization: Wildlife and Range Management
- Thesis: *Spatio-temporal analysis of landscape structure in the Ankasa Conservation Area using Landsat satellite images and field survey*
- Supervisor: Ing. Dr. George Ashiagbor

Awards and Funding

Society for Tropical Ecology Field Work Grant 2026
Competitive funding for vegetation fieldwork and remote sensing research in Ghana's forest–savanna transition zone.

Research Experience

Graduate Research Assistant 2024 – Present
University of British Columbia *Vancouver, Canada*

- Investigate long-term land-use transitions and vegetation structural change in Ghana's forest–savanna transition zone, including structural homogenization and divergence between forest and savanna systems.
- Built random forest models to classify forest, savanna, savanna woodland, transitional vegetation, and cropland, and regression models to estimate continuous tree cover from spectral, phenological, topographic, and field predictors.
- Applied PyTorch and TerraTorch to extract Prithvi foundation-model embeddings and explored Google Satellite Embeddings for historical land-cover reconstruction.
- Characterize disturbance and recovery with LandTrendr time-series segmentation, vegetation indices (NDVI, EVI, NDMI, NBR), and seasonal phenology metrics derived from Landsat imagery.
- Integrate UAV imagery, GEDI LiDAR, and vegetation plot measurements for model training, validation, and structural assessment; lead the vegetation surveys and UAV flights that produce these reference data.
- Develop reproducible processing and analysis workflows in Python, Google Earth Engine, and R.

Geospatial Analyst Aug 2023 – Dec 2024
Alliance of Bioversity International and CIAT *Accra, Ghana*

- Supported geospatial research on climate-smart agriculture, climate adaptation, and sustainable land management.
- Contributed to interactive GIS dashboard and web maps for climate information services.
- Analyzed geospatial and policy datasets on agricultural development, climate change, and environmental policy, and communicated results through reports, manuscripts, blogs, and presentations.

Research Assistant Oct 2022 – Oct 2023
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*

- Applied GIS and remote sensing to research on land-use change, carbon stocks, biodiversity conservation, and forest protected-area management using R and Python.
- Conducted spatial and statistical analyses in R for research reports and scientific publications.

Remote Sensing and GIS Technician Assistant Oct 2022 – Apr 2023
Gold Fields Ghana *Tarkwa/Damang, Ghana*

- Collected geospatial data through field surveys and satellite imagery.
- Preprocessed satellite imagery and GPS survey data for land-cover classification and environmental mapping.
- Produced aboveground and soil carbon-stock maps and contributed to technical reports.

Field Research Assistant Jan 2023 – Feb 2023
Ghana Forestry Commission / KNUST *Western Region, Ghana*

- Collected GPS-based field reference data to validate national land-cover and protected-area maps.

Research Enumerator Jul 2022 – Aug 2022
IUCN, A Rocha Ghana, World Agroforestry and partners *Ghana*

- Conducted community surveys and focus groups for a study on gender-responsive nature-based solutions for climate change and adaptation, and mapped community conservation recommendations.

Remote Sensing and GIS Intern Nov 2021 – Jan 2022
Centre for Remote Sensing and Geographic Information Services (CERGIS) *Accra, Ghana*

- Mapped land cover in the Densu catchment using Landsat imagery in Google Earth Engine, with processing and visualization in QGIS and ArcMap.

Publications and Reports

Peer-reviewed journal articles

1. Ashiagbor, G., Abubakar, S. K., **Inusah, S. S.**, Adjapong, A. O., Osei, G. N., & Laari, P. B. (2024). Analysis of the impact of agriculture and logging on forest habitat structure in the Ankasa and Bia conservation areas of Ghana. *Ecology and Evolution*, 14(12), e70712.

2. Ashiagbor, G., Quarshie, V., **Inusah, S. S.**, Essah, I. S., Abubakar, S. K., Tetteh, E. N., et al. (2024). Assessing land use change from food croplands to rubber in Ghana's Ellembelle district: Implications for food self-sufficiency. *Scientific African*, 26, e02433.

3. Dziwornu, M. G., Mponela, P., **Inusah, S. S.**, Agyarko, F. F., Yeboah, S., Damba, O. T., et al. (2025). Institutional efforts and regional distribution of climate-smart agriculture (CSA) initiatives in Ghana. *Climate Smart Agriculture*, 2(1), 100038.

4. Laryea, B., Asiamah, K., Araphat, A., Mensah, E., Tsekpo, E., Hawa, A., & **Inusah, S.** (2024). Topic modelling of the "fix the country" protest in Ghana using Latent Dirichlet Allocation (LDA) and Jaccard similarity. *Journal of Applied Science and Information Technology*, 1(1).

Reports and datasets

5. Ewell, H., Byandaga, L., Kagabo, D., Ouedraogo, M., Dalaa, M., **Inusah, S. S.**, Yeboah, S., Damba, T. O., Sartas, M., Girvetz, E., & Nzuki, E. (2024). *Ghana Scaling Week* [Report]. AICCRA, CGIAR. `https://hdl.handle.net/10568/159971`

6. Mponela, P., Dziwornu, M. G., **Inusah, S.**, Agyarko-Fosu, F., Odonkor, E. N., Sackey, T. A., & Akpatsu, I. B. (2024). *Data for the report: Towards optimising climate-smart agriculture resource investment decisions: Mapping*

program impact areas, policy support and mitigation feasibility [Dataset]. Alliance of Bioversity International and CIAT. https://hdl.handle.net/10568/138047

7. Mponela, P., Dziwornu, M. G., **Inusah, S.**, Agyarko-Fosu, F., Odonkor, E. N., et al. (2023). *Towards optimised climate-smart agriculture resource investment decisions: Mapping program impact areas, policy support and mitigation feasibility* [Technical report]. Alliance of Bioversity International and CIAT. https://hdl.handle.net/10568/136902

8. Mponela, P., Dziwornu, M. G., Agyarko-Fosu, F., **Inusah, S.**, Odonkor, E. N., Sackey, T. A., Mamah, S. A., & Akpatsu, I. B. (2023). *Climate-smart agriculture implementation evidence in Ghana: Supporting scaling strategies for enhanced resilience in Ghana* [Brief]. AICCRA Ghana Cluster Reports 2023. https://hdl.handle.net/10568/136906

Conference Presentations

Inusah, S. S. (2026, December 9). *Spatiotemporal Analysis of Vegetation Dynamics in Ghana's Forest–Savanna Transition Zone Using Multi-Source Remote Sensing* [Oral presentation, GC32A-06]. AGU Annual Meeting 2026, San Francisco, CA. Session GC32A: Advances in Remote Sensing and Modeling of Trees and Forest Ecosystems III.

Full publication list: Google Scholar.

Technical Skills

Programming	Python, R, JavaScript (Google Earth Engine)
Machine learning & GeoAI	Random Forest classification and regression, PyTorch, TerraTorch, Prithvi foundation-model embeddings, Google Satellite Embeddings
Earth observation	Landsat, Sentinel-2, HLS, GEDI LiDAR, UAV imagery; vegetation indices, phenology metrics, LandTrendr time-series segmentation
GIS software	Google Earth Engine, ArcGIS Pro, ArcGIS Online, QGIS, ArcMap, SNAP, Agisoft Metashape, FRAGSTATS
Field methods	Vegetation plot surveys, canopy-cover measurement, UAV acquisition, GPS/GNSS surveys, ecological sampling

Teaching Experience

Graduate Teaching Assistant 2024 – Present
University of British Columbia *Vancouver, Canada*

- Support undergraduate courses in GIS, environmental geography, and spatial data analysis, guiding students in ArcGIS Pro, QGIS, mapping, and spatial statistics.

University Teaching Assistant 2022 – 2023
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*

- Led practical sessions in GIS, remote sensing, forest ecosystems, and wildlife and range ecology; trained students in QGIS, ArcMap, and land-cover mapping, and wrote a practical GIS training manual.
- Supported student research projects, fieldwork, data analysis, and presentations.